ENGLISH PREMIER LEAGUE ATTACKING EVENT ANALYSIS

An Exploratory and Statistical Analysis of Attacking Involvement Using Wyscout Event Data

Capstone Project

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GitHub Repository: Michael Boateng Soccer Capstone Project

August 2026

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1. Introduction

Football has increasingly become a data-driven sport in which teams, coaches, and analysts use performance data to better understand how players and teams behave during matches. Modern football datasets record thousands of individual match events, including passes, shots, offsides, free kicks, corners, and other actions. Analyzing these events can provide useful insights into patterns of play and the involvement of teams and players in attacking situations.

Attacking performance is an important aspect of football analysis because creating offensive opportunities involves more than simply scoring goals. Players contribute to attacking play through passing, build-up actions, shooting, movement into advanced positions, and set pieces. Therefore, examining event-level data can provide a broader understanding of how attacking involvement is distributed across teams, players, positions, and event types.

This project analyzes football event data to explore patterns of attacking involvement. For this study, attacking events were operationally defined to include selected passing actions, shots, offsides, and clearly attacking set-piece situations such as corners, free-kick crosses, free-kick shots, and penalties. Generic free-kick events were excluded after exploratory analysis showed that many represented restarts by goalkeepers and defenders rather than clearly attacking actions.

The analysis focuses on the distribution of attacking events, passing patterns, positional differences, team-level involvement, and individual player involvement. Statistical analysis is also used to determine whether player position is associated with the type of attacking event performed.

It is important to distinguish **attacking involvement** from **attacking effectiveness** throughout this study. The constructed attacking-event measure is dominated by passing actions and therefore reflects participation in attacking and build-up play rather than directly measuring goal-scoring ability or attacking efficiency.

2. Project Objectives

The main objective of this project is to use event-level football data to examine patterns of attacking involvement across teams, players, positions, and event types.

The specific objectives are to:

1. Identify the distribution of the major event types included in the attacking-event dataset.
2. Examine the different types of passes contributing to attacking involvement.
3. Compare attacking-event involvement across player positions.
4. Compare attacking-event involvement across teams.
5. Identify players with the highest levels of attacking-event involvement.
6. Determine whether there is a statistically significant association between player position and attacking event type.
7. Evaluate the strength of the association between player position and attacking event type and identify the position-event combinations contributing most strongly to the relationship.

3. Research Questions

The project addresses the following research questions:

1. What types of events make up the attacking-event dataset, and how frequently does each occur?
2. Which passing sub-events are most common within the attacking-event dataset?
3. How is attacking-event involvement distributed across player positions?
4. Which teams record the highest levels of attacking-event involvement?
5. Which players record the highest levels of attacking-event involvement?
6. Is there a statistically significant association between player position and attacking event type?
7. If an association exists, how strong is it, and which player-position and attacking-event combinations contribute most strongly to the relationship?

4. Methodology and Analytical Workflow

This project followed an iterative exploratory data analysis (EDA) workflow in R. The workflow consisted of data sourcing, data understanding, data cleaning, data transformation and feature engineering, exploratory analysis, visualization, statistical analysis, and interpretation. The approach was iterative rather than strictly linear: findings from exploratory analysis were used to revisit and refine earlier data-preparation decisions when necessary.

Data sourcing: The Wyscout soccer match event dataset was obtained from the official Figshare repository linked to the peer-reviewed Scientific Data publication by Pappalardo et al. (2019).

Data understanding: The event, player, and team files were inspected to identify relevant identifiers, event categories, player roles, and team information.

Data cleaning: The datasets were joined using player and team identifiers; player-name encoding issues, duplicate records, missing values, and ambiguous event categories were investigated.

Data transformation and feature engineering: A study-specific attacking-event dataset was constructed from selected pass types, shots, offsides, and clearly attacking set pieces.

Exploratory analysis and visualization: Counts, percentages, tables, and ggplot2 visualizations were used to examine event distributions and differences across passing types, positions, teams, and players.

Statistical analysis: A Chi-Square Test of Independence assessed the relationship between player position and attacking event type. Cramér's V measured association strength, and standardized residuals identified the cells contributing most strongly to the relationship.

Interpretation: The results were interpreted in football terms while distinguishing attacking involvement from attacking effectiveness.

5. Dataset Description

5.1 Data Source and Source Publication

The data used in this project come from the public Wyscout soccer match event dataset released with the peer-reviewed article by Pappalardo et al. (2019), *A public data set of spatio-temporal match events in soccer competitions*. The publication describes event data collected by Wyscout for seven soccer competitions, including the 2017/2018 English first division, and reports that the collection contains event, player, team, match, referee, and coach information. The dataset is publicly available through the official Figshare repository and is released under a CC BY 4.0 license.

Peer-reviewed publication: https://doi.org/10.1038/s41597-019-0247-7

Official dataset repository: https://doi.org/10.6084/m9.figshare.c.4415000

The project uses event-level football data containing detailed information about actions performed during matches. The analysis combines event, player, and team information to examine attacking involvement at different levels.

The event data contain information such as the *event type, sub-event type, player identifier, team identifier, match identifier, match period,* and *event time*. Examples of recorded events include passes, shots, free kicks, and offsides.

Player information was incorporated into the event data using the player identifier. This provided additional variables such as player name and playing position. The positions used in the analysis were:

- Midfielder
- Defender
- Forward
- Goalkeeper

Team information was also joined to the event-level data using the team identifier, allowing events to be associated with their respective teams.

The primary variables used in the analysis included *eventName*, *subEventName, playerId, teamId, matchId, eventSec, shortName, role_name,* and *team_name.*

During initial data inspection, it was observed that the *event_players* field contained **48,031 missing player names**. Further investigation showed that these missing values were not random but corresponded to **interruptions in play**, such as stoppages where no identifiable player was recorded. As a result, these observations were not treated as data quality errors but as a meaningful feature of the dataset structure.

Additionally, the *eventSec* variable was examined in detail. It was found that the dataset included events recorded up to approximately **58 minutes of match time** in certain cases. This prompted further investigation into the temporal structure of the data. It was confirmed that this did not indicate missing match data but rather reflected the segmentation and recording conventions used in the event-tracking system.

After data preparation and the construction of the attacking-event measure, the final dataset contained **342,225 attacking-event observations**.

6. Data Cleaning and Preparation

6.1 Combining the Datasets

The event data were joined with the player dataset using the player identifier. This allowed each event to be associated with available player information, including the player's name and playing position.

Team information was subsequently joined using the team identifier, allowing attacking events to be analyzed at the team level.

6.2 Player Name Cleaning

During data inspection, several player names contained Unicode escape sequences rather than correctly displayed accented characters. Examples included names such as N. Kanté, Fàbregas, N. Matić, M. Özil, and A. Doucouré.

The affected character strings were decoded so that player names were displayed correctly. This improved the readability and consistency of player-level tables and visualizations.

6.3 Duplicate Records

The final attacking-event dataset was examined for completely duplicated observations. No completely duplicated records were identified.

Therefore, no observations were removed on the basis of complete duplication.

6.4 Missing Values

Missing-value checks were conducted on the variables used in the analysis. No missing observations were identified for event name, sub-event name, player identifier, team identifier, team name, or event time.

However, **39 observations** were missing, both player name and positional information. These represented approximately **0.01%** of the **342,225** attacking-event observations.

Rather than removing these observations from the entire dataset, they were retained for analyses that did not require player-level information. They were excluded only from analyses involving player identity or player position.

6.5 Construction of the Attacking-Event Dataset

For the purpose of this study, an operational definition of attacking involvement was created from selected event categories.

The final attacking-event dataset included the following passing actions:

- Simple pass
- High pass
- Head pass
- Cross
- Launch
- Smart pass

Shots and offsides were also included.

Attacking set pieces consisted of:

- Corners
- Free-kick crosses
- Free-kick shots
- Penalties

During exploratory analysis, generic free-kick events were initially included in the attacking-event definition. Further investigation showed that many of these events represented restarts taken by goalkeepers and defenders and therefore could not reliably be classified as attacking actions.

Generic free kicks were consequently excluded, while corners, free-kick crosses, free-kick shots, and penalties were retained because they represented more clearly identifiable attacking set-piece situations.

Following this refinement, the attacking-event dataset contained **342,225 observations**.

The final attacking-event classification was checked to ensure that only the intended event and sub-event categories remained.

The dataset contained:

- **326,183 passes**
- **8,451 shots**
- **6,033 attacking set-piece events**
- **1,558 offsides**

These categories produced the final total of:
326,183 + 8,451 + 6,033 + 1,558 = 342,225

Additional checks confirmed that the generic free-kick category was no longer present in the final attacking-event dataset.

These cleaning and validation procedures ensured that the subsequent exploratory and statistical analyses were conducted using a consistent operational definition of attacking involvement.

7. Exploratory Data Analysis and Results

Exploratory Data Analysis (EDA) was conducted to understand the distribution and patterns of the events included in the final attacking-event dataset. The analysis examined overall event types, passing sub-events, player positions, teams, and individual players.

7.1 Overall Attacking-Event Distribution

The final attacking-event dataset contained **342,225 observations**. Passing represented the largest category by a substantial margin.

Event Type	Number of Events	Percentage
Pass	326,183	95.31%
Shot	8,451	2.47%
Free Kick	6,033	1.76%
Offside	1,558	0.46%

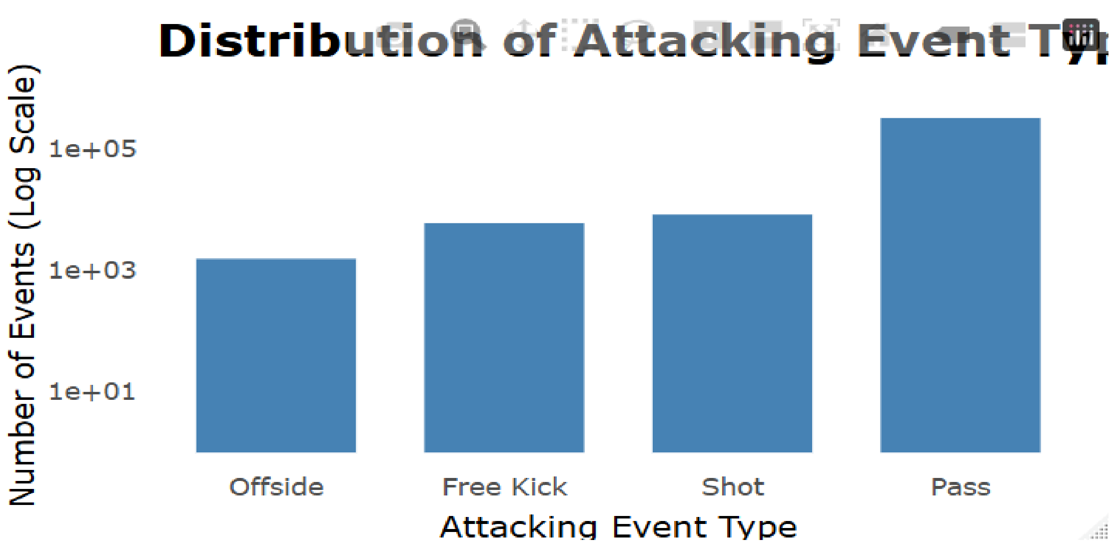

Figure 1. Distribution of selected attacking event types.

Passes accounted for **95.31%** of all events included in the attacking-event dataset. Shots accounted for 2.47%, attacking set pieces categorized under Free Kick accounted for 1.76%, and offsides represented 0.46%.

The dominance of passing demonstrates that the operational measure used in this study primarily captures involvement in passing and build-up play. Therefore, these percentages should not be interpreted as indicating that 95.31% of football attacks were passes. A single attacking sequence may contain multiple passes before a shot or other attacking outcome occurs.

7.2 Passing Sub-Event Analysis

Because passing represented the majority of the attacking-event dataset, the passing category was examined in greater detail.

A total of **326,183 passing events** were included in the analysis. These consisted of six passing sub-events:

Passing Type	Number of Events
Simple pass	251,405
High pass	25,067
Head pass	21,332
Cross	12,251
Launch	10,247
Smart pass	5,881

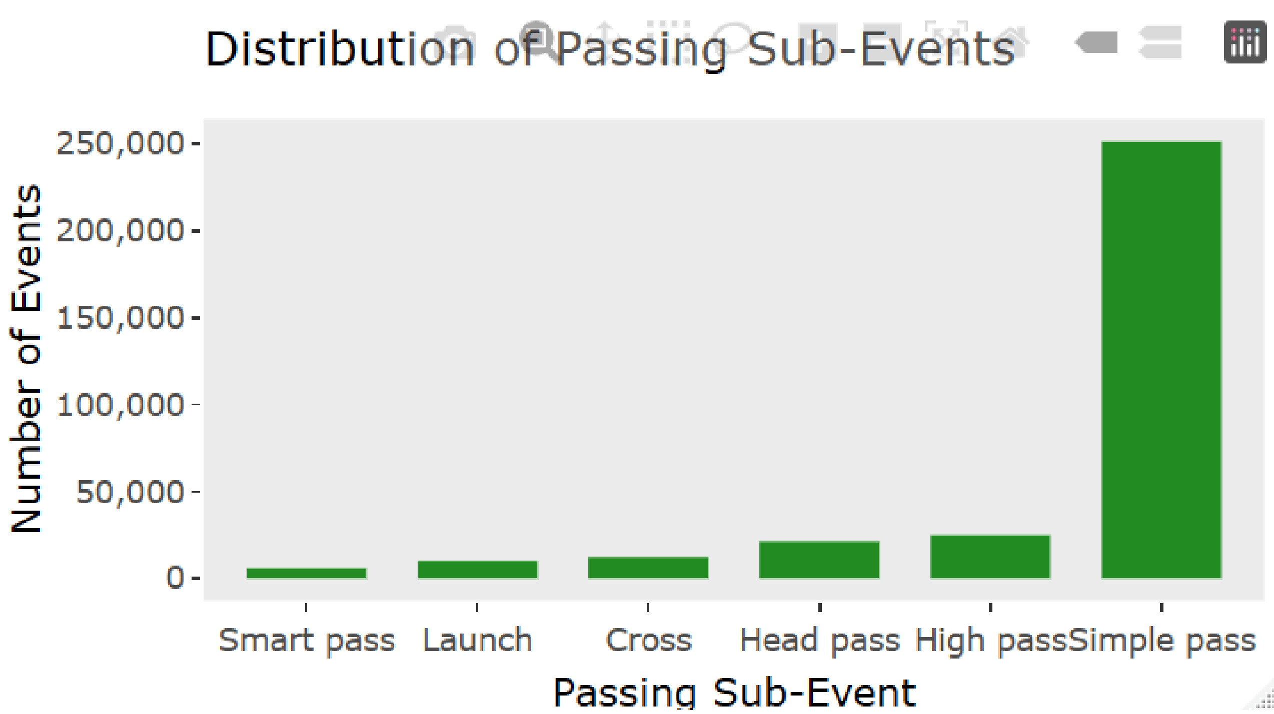

Figure 2. Distribution of passing sub-events in the attacking-event dataset.

Simple passes were by far the most frequently recorded passing action, accounting for approximately **77.1%** of the selected passes. High passes and head passes were the next most frequent categories, while smart passes were the least common.

The results indicate that routine ball circulation and possession actions constituted a substantial part of the attacking involvement captured by the dataset. More specialized actions, such as crosses and smart passes, occurred much less frequently.

This finding is important when interpreting later team and player rankings because players who are heavily involved in passing and build-up play will naturally accumulate more events under this definition.

7.3 Attacking Involvement by Player Position

Attacking-event involvement was next examined according to player position. Observations with missing positional information were excluded from this specific analysis.

Position	Number of Events	Percentage
Midfielder	148,643	43.44%
Defender	140,899	41.18%
Forward	41,014	11.99%
Goalkeeper	11,630	3.40%

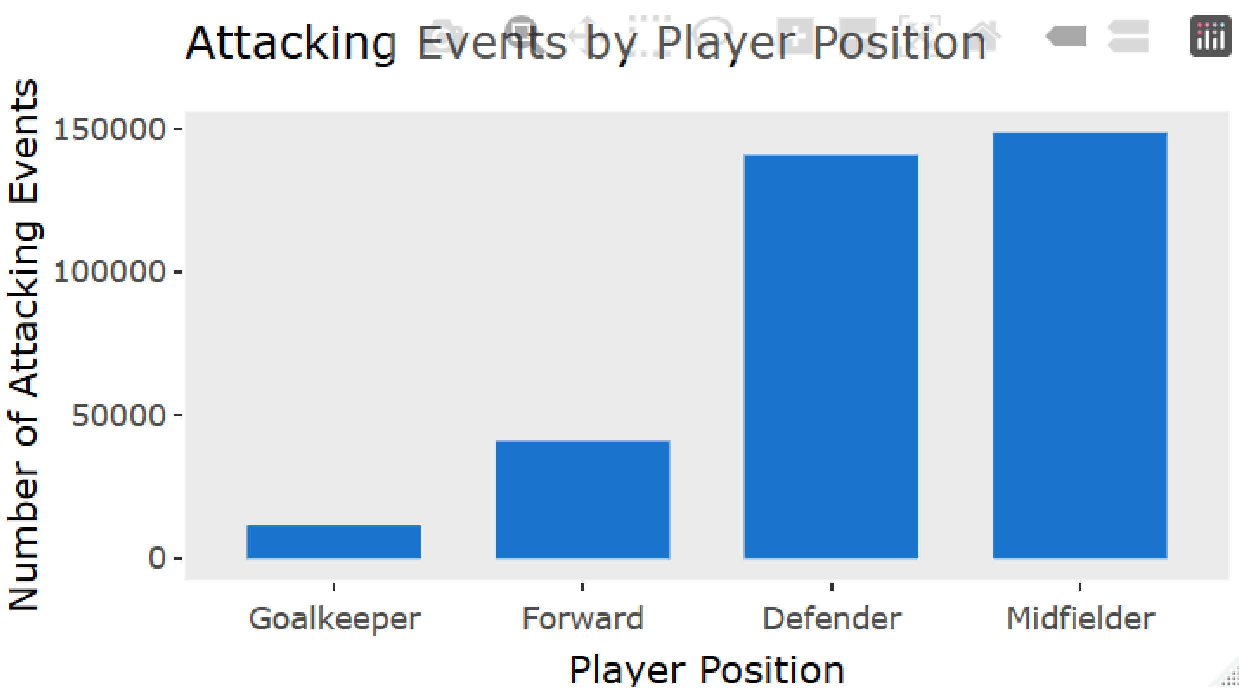

Figure 3. Attacking-event involvement by player position.

Midfielders recorded the largest share of attacking events at **43.44%**, followed closely by defenders at **41.18%**. Forwards accounted for 11.99%, while goalkeepers accounted for 3.40%.

At first glance, the high involvement of defenders relative to forwards may appear surprising. However, this pattern is largely explained by the dominance of passing events in the constructed measure. Defenders frequently participate in possession and build-up phases and therefore accumulate large numbers of passing events.

Consequently, these results should not be interpreted as evidence that defenders are more effective attackers than forwards. Rather, they indicate that midfielders and defenders had substantial involvement in the selected actions used to represent attacking and build-up participation.

7.4 Attacking Involvement by Team

Team-level analysis revealed variation in the total number of attacking events recorded by the 20 teams.

The six teams with the highest attacking-event counts were:

Rank	Team	Attacking Events	Percentage
1	Manchester City	28,420	8.30%
2	Arsenal	23,506	6.87%
3	Liverpool	23,067	6.74%
4	Tottenham Hotspur	21,835	6.38%
5	Chelsea	21,255	6.21%
6	Manchester United	20,097	5.87%

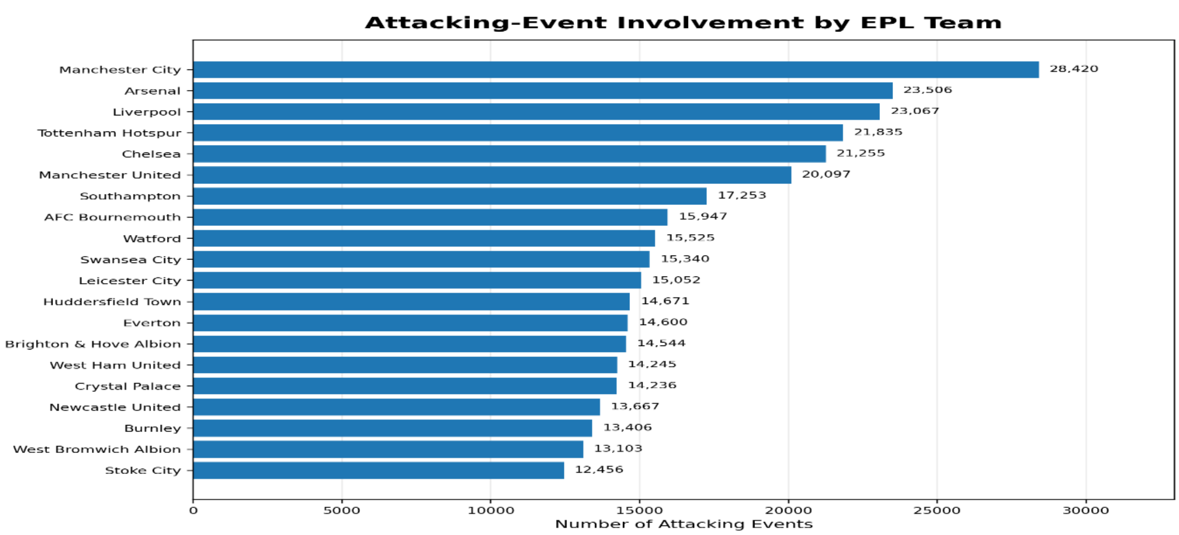

Figure 4. Attacking-event involvement across the 20 English Premier League teams.

Manchester City recorded the highest attacking-event involvement, with **28,420 events (8.30%)**. Arsenal ranked second with 23,506 events (6.87%), followed closely by Liverpool with 23,067 (6.74%).

Tottenham Hotspur, Chelsea, and Manchester United completed the top six.

Manchester City's total was noticeably higher than those of the other teams, suggesting greater involvement in the selected attacking and build-up actions represented by the dataset.

However, because passing constitutes 95.31% of the constructed attacking-event measure, these rankings should not be interpreted directly as rankings of attacking effectiveness. Instead, they represent the volume of involvement in the selected attacking and build-up events.

7.5 Player-Level Attacking Involvement

Individual player involvement was examined by counting the number of attacking events associated with each player.

The highest-ranked players were:

Rank	Player	Team	Position	Attacking Events
1	G. Xhaka	Arsenal	Midfielder	3,151
2	N. Otamendi	Manchester City	Defender	2,995
3	K. De Bruyne	Manchester City	Midfielder	2,958
4	Fernandinho	Manchester City	Midfielder	2,891
5	Azpilicueta	Chelsea	Defender	2,726
6	David Silva	Manchester City	Midfielder	2,493
7	N. Matić	Manchester United	Midfielder	2,477
8	C. Eriksen	Tottenham Hotspur	Midfielder	2,476
9	J. Vertonghen	Tottenham Hotspur	Defender	2,394
10	K. Walker	Manchester City	Defender	2,331
11	Fàbregas	Chelsea	Midfielder	2,242
12	A. Doucouré	Watford	Midfielder	2,225
13	E. Dier	Tottenham Hotspur	Midfielder	2,178
14	N. Kanté	Chelsea	Midfielder	2,126
15	J. Henderson	Liverpool	Midfielder	2,014
16	A. Mooy	Huddersfield Town	Midfielder	2,011
17	M. Özil	Arsenal	Midfielder	1,934
18	Bellerín	Arsenal	Defender	1,909
19	D. Sánchez	Tottenham Hotspur	Defender	1,809
20	Oriol Romeu	Southampton	Midfielder	1,782

G. Xhaka recorded the highest attacking-event involvement with **3,151 events**, followed by N. Otamendi with 2,995 and K. De Bruyne with 2,958.

Midfielders and defenders dominated the top 20, while no forwards appeared among the highest-ranked players. Manchester City was particularly well represented, with several players appearing near the top of the ranking.

This pattern is consistent with the earlier positional analysis. Since passing accounts for the overwhelming majority of the attacking-event dataset, midfielders and defenders who participate extensively in ball circulation and build-up play accumulate large event totals.

Therefore, the player rankings should be interpreted as measures of **attacking and build-up involvement under the operational definition used in this study**, rather than as rankings of the best or most effective attacking players.

Overall, the exploratory analysis demonstrates substantial differences in attacking-event involvement across event types, positions, teams, and players. It also highlights the importance of considering the composition of the attacking-event measure when interpreting raw event counts.

8. Statistical Analysis

Following the exploratory analysis, inferential statistical methods were used to determine whether the type of attacking event performed was associated with player position.

8.1 Chi-Square Test of Independence

A Pearson Chi-Square Test of Independence was conducted to examine the relationship between **player position** and **attacking event type**.

The hypotheses were:

H_0: Player position and attacking event type are independent.
H_1: Player position and attacking event type are associated.

The test produced $\chi^2(9) = 14{,}592$, $p < 2.2 \times 10^{-16}$.

Pearson's Chi-squared test

data: position_event_table

X-squared = 14592, df = 9, p-value < 2.2e-16

Since the p-value is substantially below the conventional significance level of 0.05 ($p < 2.2 \times 10^{-16}$), the null hypothesis is rejected.

Therefore, there was statistically significant evidence of an association between **player position and attacking event type**.

However, statistical significance alone does not indicate how strong the relationship is. Because the dataset contains a very large number of observations, relatively small differences may produce statistically significant results. Therefore, Cramér's V was also calculated as a measure of association strength.

8.2 Strength of Association: Cramér's V

Cramér's V produced:

0.1192

This value indicates that although player position and attacking event type were statistically associated, the **overall strength of the association was relatively modest**.

Therefore, the extremely small p-value should not be interpreted as evidence of a very strong relationship. Instead, the results indicate that event-type distributions differ systematically across positions, while considerable variation remains within each position.

8.3 Standardized Residual Analysis

To identify which position-event combinations contributed most strongly to the significant Chi-Square result, standardized residuals were examined.

Player Position	Free Kick	Offside	Pass	Shot
Defender	-45.59	-24.55	69.59	-45.49
Forward	-7.24	65.47	-74.15	78.66
Goalkeeper	-13.51	-7.41	23.59	-17.46
Midfielder	54.95	-15.82	-29.13	0.01

Positive standardized residuals indicate that a position-event combination occurred **more frequently than expected** under independence, while negative values indicate that it occurred **less frequently than expected**.

Residuals with large absolute values provide evidence that a particular cell contributes substantially to the overall Chi-Square association.

Defenders

Defenders recorded substantially more passes than expected, with a standardized residual of **69.59**. In contrast, their free kicks, offsides, and shots occurred less frequently than expected.

This pattern is consistent with defenders' substantial involvement in ball circulation and build-up play.

Forwards

Forwards demonstrated the strongest positive residual for shots, **78.66**. They also recorded substantially more offsides than expected, **65.47**

At the same time, forwards recorded substantially fewer passes than expected, with a standardized residual of −74.15.

These findings are consistent with forwards being more heavily involved in advanced attacking actions, including attempts on goal and movements that may result in offside decisions.

Goalkeepers

Goalkeepers recorded more passes than expected, with a standardized residual of **23.59**, while free kicks, offsides, and shots occurred less frequently than expected within the selected attacking-event categories.

This result is notably different from the preliminary analysis in which generic free kicks were included. Removing generic restart events produced a more interpretable pattern of goalkeeper involvement.

Midfielders

Midfielders recorded substantially more attacking set-piece events than expected, with a standardized residual of **54.95**.

Their shot residual was approximately zero (0.01), indicating that the observed number of midfielder shots was very close to what would be expected under the independence model.

Midfielders also recorded fewer passes than expected relative to the distribution predicted by the model. This does **not** mean that midfielders recorded few passes in absolute terms. Instead, it means that, given their overall event volume and the marginal distribution of event types, their share of passes was lower than expected under independence.

8.4 Statistical Conclusion

The statistical analysis provides evidence that player position is associated with attacking event type: $\chi^2(9) = 14{,}592$, $p < 2.2 \times 10^{-16}$.

However, Cramér's V = 0.1192 shows that the overall strength of this relationship is relatively modest.

The standardized residuals reveal more specific positional patterns. Defenders and goalkeepers were disproportionately represented in passing events, forwards were disproportionately represented in shots and offsides, and midfielders were disproportionately represented in attacking set-piece events.

These results support the conclusion that different playing positions exhibit distinct patterns of involvement across the selected attacking-event categories, although player position alone does not strongly determine the type of attacking event performed.

9. Discussion

The purpose of this project was to explore patterns of attacking involvement using event-level football data and to determine how these patterns varied across event types, player positions, teams, and individual players.

One of the clearest findings was the dominance of passing within the constructed attacking-event dataset. Passes accounted for **95.31%** of all selected attacking events. This indicates that the operational definition used in this study captures a substantial amount of build-up and ball-circulation activity. Consequently, total attacking-event counts should primarily be viewed as measures of involvement in attacking and build-up play rather than direct measures of attacking effectiveness.

This helps explain the positional findings. Midfielders accounted for the largest share of attacking events (**43.44%**), followed closely by defenders (**41.18%**). Forwards accounted for only 11.99%. These results do not imply that defenders are more effective attackers than forwards. Rather, midfielders and defenders accumulate substantial event counts because of their involvement in passing and build-up actions.

The statistical analysis provides further insight into these positional differences. The Chi-Square Test of Independence showed a statistically significant association between player position and attacking event type, $\chi^2(9) = 14{,}592$, $p < 2.2 \times 10^{-16}$. However, Cramér's V = 0.1192 showed that the overall association was relatively modest.

The standardized residuals revealed more specific patterns that were not apparent from the overall event totals alone. Defenders recorded more passes than expected, while forwards recorded substantially more shots and offsides than expected. Goalkeepers were also more strongly represented in passing, while midfielders were more strongly represented in attacking set pieces.

These findings demonstrate why event totals should be interpreted together with event type. Although forwards accumulated fewer total events, their disproportionately high involvement in shots and offsides reflects a different attacking role from that of midfielders and defenders.

At the team level, **Manchester City** recorded the highest attacking-event involvement with 28,420 events (8.30%), followed by Arsenal and Liverpool. Manchester City also had several players among the highest-ranked individuals, including N. Otamendi, K. De Bruyne, Fernandinho, David Silva, and K. Walker. This suggests substantial involvement in the selected passing, build-up, and attacking actions across multiple players.

At the individual level, G. Xhaka recorded the highest attacking-event total, followed by N. Otamendi and K. De Bruyne. Midfielders and defenders dominated the top-player rankings. Again, this pattern should be interpreted in the context of the high proportion of passes in the attacking-event measure rather than as a ranking of the most effective attacking players.

Overall, the findings demonstrate that attacking involvement is multidimensional. Total event volume captures one aspect of participation, while the types of events performed provide additional information about the different tactical roles played by teams, positions, and individual players.

10. Limitations

Several limitations should be considered when interpreting the results of this study.

First, the operational definition of attacking involvement is heavily influenced by passing. Passes account for **95.31%** of the final attacking-event dataset. As a result, teams and players with greater involvement in passing and build-up play naturally accumulate higher attacking-event totals.

Second, the analysis measures **attacking involvement rather than attacking effectiveness**. A player recording a large number of attacking events is not necessarily a more effective attacker than a player recording fewer events. The current analysis does not directly evaluate measures such as goal conversion, chance quality, expected goals, assists, or attacking efficiency.

Third, individual events were analyzed primarily as separate observations. Football attacks, however, often consist of sequences of connected actions. A series of passes, for example, may eventually lead to a shot. Counting each event separately does not fully capture these possession and attacking sequences.

Fourth, **39 observations** were missing player names and positional information. Although this represented only approximately 0.01% of the final dataset and therefore had minimal impact on the overall analysis, these observations could not be included in analyses requiring player identity or position.

Finally, the results are specific to the teams, players, matches, and event definitions represented in the dataset. Care should therefore be taken when generalizing the findings beyond the context analyzed in this project.

11. Recommendations and Future Work

Future analysis could extend this project by moving from measures of attacking involvement toward measures of attacking effectiveness.

One useful extension would be to analyze **shot conversion**, comparing the number of shots taken with the number of goals scored. This would allow teams and players to be evaluated not only by attacking-event volume but also by how efficiently attacking opportunities are converted into goals.

Another extension would involve reconstructing **attacking or possession sequences**. Rather than analyzing individual events independently, events could be connected chronologically within matches to examine how passes and other actions contribute to the creation of shooting opportunities.

Spatial information could also be incorporated to determine where passes and shots occur on the pitch. This could help distinguish routine possession in deeper areas from actions performed in more dangerous attacking areas.

Future studies could additionally incorporate measures such as expected goals, assists, key passes, shot locations, possession length, or other efficiency-based metrics where such information is available.

These extensions would provide a more complete assessment of attacking effectiveness while building on the event-level patterns identified in the present study.

12. Conclusion

This project used event-level football data to investigate attacking involvement across event types, player positions, teams, and individual players.

The final attacking-event dataset contained **342,225 observations**, with passing accounting for **95.31%** of the selected events. Midfielders recorded the largest share of attacking-event involvement at **43.44%**, closely followed by defenders at **41.18%**.

At the team level, Manchester City recorded the highest attacking-event involvement with **28,420 events (8.30%)**, followed by Arsenal and Liverpool. At the individual-player level, G. Xhaka recorded the highest number of attacking events, followed by N. Otamendi and K. De Bruyne.

Inferential analysis showed a statistically significant association between player position and attacking event type, (X-squared = 14592, df = 9, p-value < 2.2e-16). However, Cramér's (V=0.1192) indicated that the overall strength of this association was relatively modest. Standardized residual analysis showed that defenders and goalkeepers were more strongly represented in passing, forwards in shots and offsides, and midfielders in attacking set-piece events.

The findings demonstrate that different player positions contribute to attacking play in different ways. They also show that high event volume should not automatically be interpreted as greater attacking effectiveness, particularly because passing dominates the operational measure used in this study.

Overall, the project demonstrates how exploratory data analysis and statistical methods can be combined to transform large event-level football datasets into interpretable insights about team, player, and positional involvement. A more complete assessment of attacking effectiveness would require additional outcome- and efficiency-based measures alongside the involvement measures examined in this study.

13. References

Pappalardo, L., Cintia, P., Rossi, A., Massucco, E., Ferragina, P., Pedreschi, D., & Giannotti, F. (2019). *A public data set of spatio-temporal match events in soccer competitions.* Scientific Data, 6, 236. https://doi.org/10.1038/s41597-019-0247-7

Pappalardo, L., & Massucco, E. (2019). *Soccer match event dataset* [Data set]. Figshare. https://doi.org/10.6084/m9.figshare.c.4415000

14. Appendices

Appendix A: Reproducibility and R Project Files

All data preparation, cleaning, exploratory data analysis, visualization, and statistical analysis for this project were conducted using **R**. The complete R scripts, generated figures, summary tables, and project documentation are available through the project's GitHub repository.

GitHub Repository: Michael Boateng Soccer Capstone Project

The analysis was organized into five R scripts that should be executed sequentially.

Script	Purpose
00_Load_Packages.R	Loads the R packages required for the project.
01_Import_Data.R	Imports the Wyscout event, player, and team JSON files.
02_Clean_Data.R	Performs data-quality checks, cleans player names, joins datasets, and constructs the final attacking-event dataset.
03_EDA.R	Performs exploratory data analysis and generates summary tables.
04_Visualization_Statistics.R	Produces visualizations and performs the Chi-Square test, Cramér's V analysis, and standardized residual analysis.

The principal R packages used were `tidyverse`, `jsonlite`, `skimr`, `stringi`, `rcompanion`, and `plotly`.

Appendix B: Operational Definition of Attacking Events

For this study, attacking involvement was operationally defined using selected event categories and sub-events from the Wyscout dataset.

Event Category	Included Sub-Events
Pass	Simple pass, Smart pass, Cross, High pass, Head pass, Launch
Shot	Shot
Offside	Offside
Free Kick	Corner, Free kick cross, Free kick shot, Penalty

Generic free-kick events that did not correspond to these selected attacking set-piece categories were excluded. This prevented routine restart events from being classified as attacking involvement.

After applying the operational definition, the final attacking-event dataset contained **342,225 observations**.

Appendix C: Final Data-Quality Assessment

Final data-quality checks were performed before conducting the exploratory and statistical analyses.

No duplicate observations were identified in the final attacking-event dataset.

The variables `eventName`, `subEventName`, `playerId`, `teamId`, `team_name`, and `eventSec` contained no missing values.

A total of **39 observations** had missing `shortName` and `role_name` information. These observations were retained when player identity or position was not required and excluded from analyses that specifically required these variables.

Unicode escape sequences in player names were also corrected during data cleaning. This allowed names such as **N. Kanté, Fàbregas, A. Doucouré, N. Matić, and M. Özil** to be represented correctly.

Appendix D: Statistical Analysis

A Pearson Chi-Square Test of Independence was conducted to determine whether player position was associated with attacking event type.

The test produced $\chi^2(9) = 14{,}592$, $p < 2.2 \times 10^{-16}$.

The null hypothesis of independence was therefore rejected, providing evidence of a statistically significant association between player position and attacking event type.

Cramér's V was 0.1192.

This indicated that although the association was statistically significant, its overall strength was relatively modest.

The standardized residuals were:

Player Position	Free Kick	Offside	Pass	Shot
Defender	-45.59	-24.55	69.59	-45.49
Forward	-7.24	65.47	-74.15	78.66
Goalkeeper	-13.51	-7.41	23.59	-17.46
Midfielder	54.95	-15.82	-29.13	0.01

Positive standardized residuals indicate position-event combinations occurring more frequently than expected under independence, whereas negative residuals indicate combinations occurring less frequently than expected.

Appendix E: Raw Data Availability

The analysis used three files from the Wyscout Soccer Match Event Dataset:

- `events_England.json`
- `players.json`
- `teams.json`

The raw JSON files are not included in the GitHub repository because events_England.json (approximately 180 MB) exceeds GitHub's standard individual file-size limit.

For local reproduction of the analysis, the three raw files should be placed in:

```
data/raw/
```

The import script is configured to read the files from this directory.

The generated figures and summary tables are provided in the figures/ and tables/ directories of the GitHub repository.